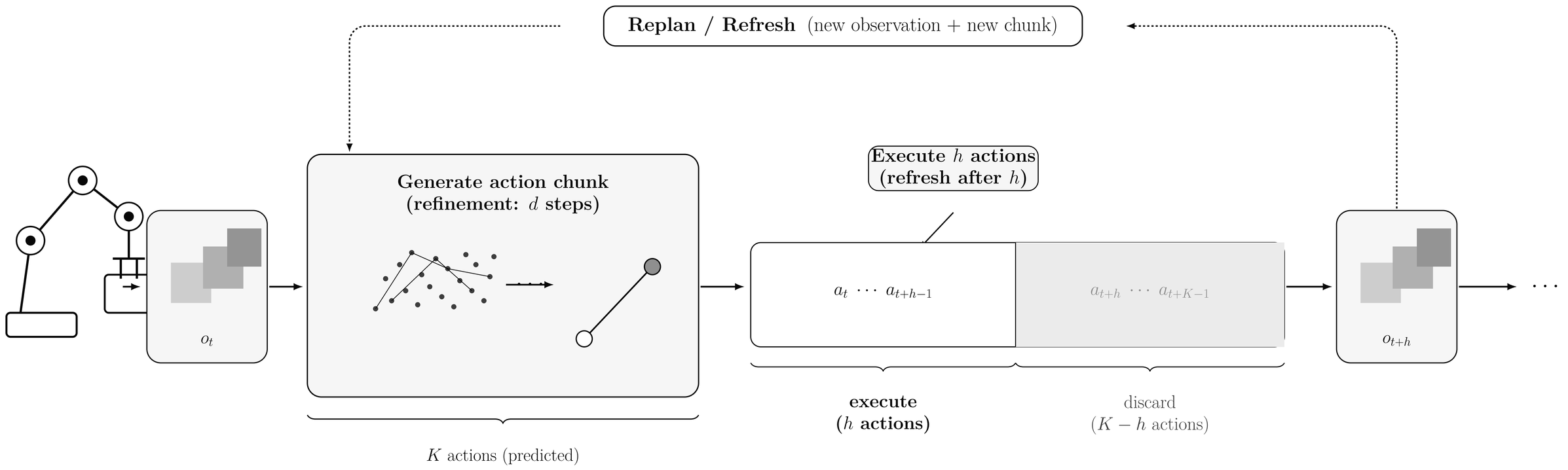
Refinement vs. Refresh

Compute Allocation for Closed-Loop Robot Policies

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Chunked Closed-Loop Policy



K: **chunk length**
(total predicted actions)

d: **refinement**
(denoising steps)

h: **refresh**
(actions before replanning)

Setup (held-out)

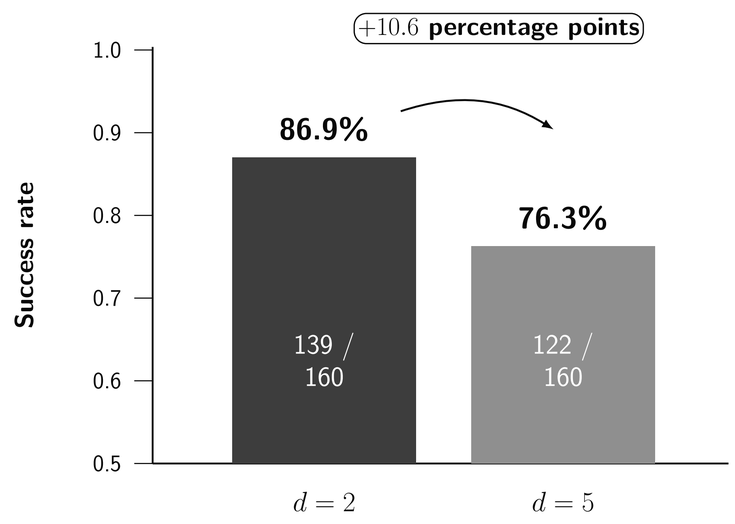
SmolVLA (frozen) $K = 50,\ d \in \{2, 5\}$

LIBERO (simulation) $h \in \{2, 5\}$ 8 tasks, 1,120 attempts

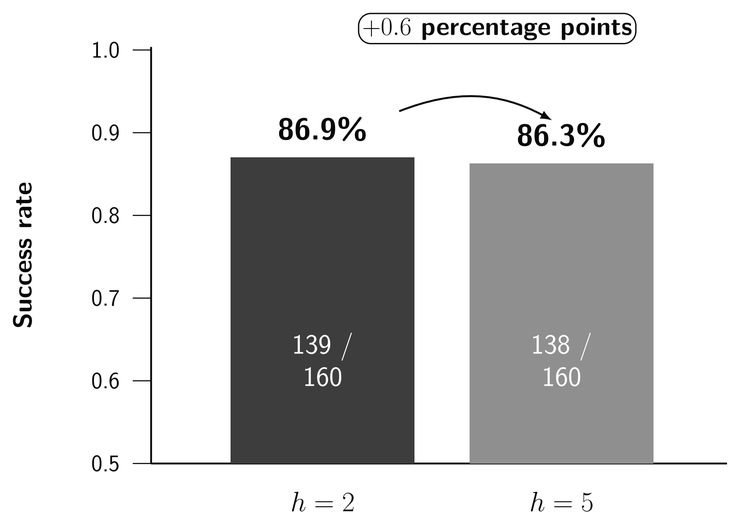
?

Under a fixed inference budget, should we allocate compute to additional denoising steps (**refinement**) or to more frequent observation-conditioned replanning (**refresh**)?

Result 1: Fewer Denoising Steps Improve Success



Result 2: Less Frequent Refresh Is Much Cheaper



What the results mean

Takeaway

At fixed $h = 2$, reducing denoising from 5 to 2 steps raises success by 10.6 percentage points and uses 5.9% less arithmetic per control step. At $d = 2$, executing $h = 5$ rather than $h = 2$ uses 60.7% less arithmetic per attempt.

Interpretation and limits

The $h = 5$ comparison gives 138/160 versus 139/160 successes, so similar quality is descriptive rather than established. Results are from one frozen checkpoint in synchronous LIBERO; arithmetic is not deployment latency, and transfer remains untested.